Paper Reliability - Title TBD

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Abstract

hello, this is the abstract.

Keywords: quadruple exit, polarity, WGM

1. Introduction

Se le da el trasfondo breve a la situacion Es irreal pensar que se deben hacer misiones de nano o smallsats que duren +7 años. Estos sistemas deben de ser reemplazados periodicamente Definir BREVEMENTE la relevancia del cubesat y por qué duran tan poco Yes, this is the introduction.[1]

The goal of this step is to estimate a system lifecycle RoM cost for the space segment of a CubeSat based DSM providing data over the relevant period (considering different deployment strategies and redundancies to achieve set reliability). A relevant value comes from the assessment of Planet, which is the most relevant analogy as its payload is a telescope, in Section 3.1, which sets an LCC cost (ideal) which includes replicability and learning is 600 k USD for a 3 U. Note, from the discussion that planet had to perform significant investments in their ground segment (this is not likely to be the case for a new system being deployed as extensive ground segment providers have deployed lower cost options).

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[1]This is the first author footnote.

[2]Another author footnote, this is a very long footnote and it should be a really long footnote. But this footnote is not yet sufficiently long enough to make two lines of footnote text.

[3]Yet another author footnote.

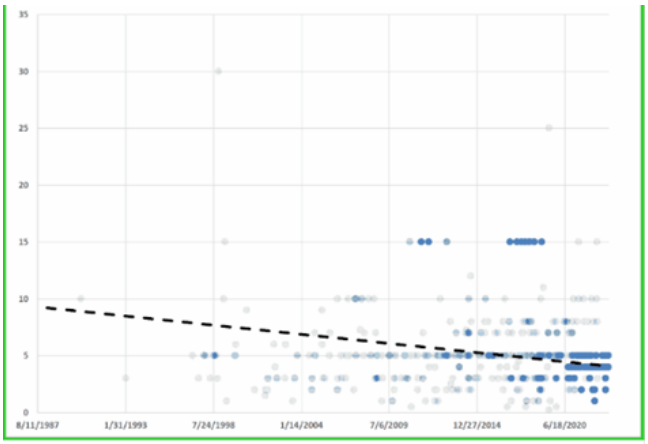

Figure 1: Results in Physics Journal cover

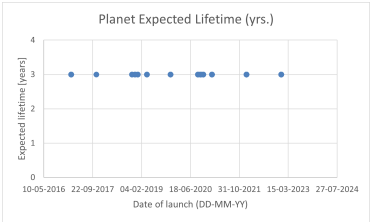

Figure 2: Results in Physics Journal cover

1.1. CubeSat LEO Lifetime

Regarding the design lifetime of CubeSats and small satellites in LEO, data from the Union of Concerned Scientists (UCS, 2021) shows a decreasing tendency in the expected (design) lifetime for LEO satellites at 4 years see Figure 1.

For the developer surveys and the classification data, a lower design lifetime for CubeSats is observed, with the biggest developer Planet declaring 3 years since 2016 to 2023 (latest dataset available), see Figure 2.

Thus, is reasonable to assume a scenario where design lifetime for LEO CubeSats is in the order of magnitude of 3 years. Note this will have a significant impact on the deployment strategy.

1.2. Number of satellites

From the past analysis, we know that 8 satellites are required in an SSO orbit at 600 km with a sensor that has a spatial resolution of 50 m to re visit the AoI twice a day. Considering the lifetime of 3 years in the past section, and the mission lifetime of 8 years, this yields a minimum of 3 generation of

satellites. This translates into

$$3 \times 8 = 24$$

satellites. The next step looks further into details regarding reliability and deployment to minimize the number of satellites (this is the costlier element) and follow a launch strategy that take advantage of "bulk" launches to lower the deployment costs.

Costs for ground segment, based on scenario 11 with 8 operational satellites produces in the order of 100000 Mbit per day (see Figure 5 32) at 6 Mbps

$$\frac{100000Mbit}{6Mbps} = 16.67e3\,[s] = 227[min]$$

Factor by 1/0.8 to get the communications time.

$$\frac{227\,[min]}{0.8} = 347\,[min] \approx 6[hrs]$$

So, we have about 6 hours of ground station time per day in order to download all the data generated by the eight satellites we need to have operational in order to fulfil the temporal resolution requirements. At 3 to 5 euros a min for S-band capable ground segment, we get an estimated.

$$6\left[\frac{hrs}{day}\right] * 3\left[\frac{Euros}{min}\right] * 365\left[\frac{day}{year}\right] * 8\,[years] = 3.156\ [MEuros]$$

This is 400 k Euros per year of operational mission. With these values, one could easily cover the costs of deploying a new ground station and converting the salaries.

1.3. Context Reliability

The following section documents the methodology implemented for assessing the reliability of a distributed spacecraft mission (DSM) based on CubeSats in LEO for different deployment strategies. The algorithm and tool for these evaluations was implemented in Python and is available at the thesis repository.

An algorithm quantifies the reliability over time of different scenarios for can a distributed spacecraft mission (DSM). The implementation requires a series of inputs, see Table 5 13 , and processes the data following the approach described in Section 4.2, to provide outputs in the form of a graph of reliability over time for the different scenarios, and a set of tables for reliability every six months together with cost estimates for the different launch strategies of the replenishment spacecraft over the mission lifetime (see Figure 5 33 for an overview).

The algorithm was developed in parallel to the case study dealt in Chapter 5. As so, many of the values used are referred to the mission, such as lifetime and reliability. The deployment cost is approached through the brokered launch of POD or using the Space-X small satellite rideshare program, both for an SSO orbit.

2. CubeSat Reliability Model

CubeSat reliability literature suggests increasing testing time (Faure et al. 2017) instead of subsystems redundancy to improve reliability (Bouwmeester, Menicucci, Gill, 2022b). Recognizing the value of increased testing considering the launch costs and the satellite costs, the approach followed in this implementation is to periodically deploy "new" satellites in orbit to keep the distributed system reliability above the required threshold. By replenishing the cluster regularly, through a certain deployment strategy, one can assess the number of satellites required over time to fulfil the requirements (even at lower individual element reliability). This is further developed in the next pages. We use the reliability of a CubeSat over time fit by Bouwmeester et al (2022), as the team at Delft University of Technology has been publishing a series of analysis on CubeSat and small satellite reliability and the latter article presents one of the few applicable models to CubeSats (which are the scope of this research). The article establishes that the Lognormal-Gompertz was the most adequate fit to the existing data at CubeSat level, while at subsystem level the Bayesian interference. Thus, the algorithm implements this as the reliability function over time. Both are shown in Figure 5 34.

The relationship for the subsystems of a CubeSat in series is defined in Eq. 5 9, where R_{sys} represents the reliability of the CubeSat and $R_{subsys,i}$ represents the reliability of each subsystem in series.

$$R_{sat}\left(t\right) = \prod_{i=1}^{n} R_{subsys,i}(t)$$

An example of the sequence of subsystems in series is presented in Figure 5 35.

At the same time the model of the reliability obtained by the authors corresponds to a Lognormal-Gompertz product mixture, shown in Eq. 5 10.

$$R_{subsys}\left(t\right) = \; e^{-\eta\left(e^{\frac{t}{\theta}}-1\right)} \;\; \frac{1}{\sigma\sqrt{2\pi}} \int_0^t \frac{2}{x} e^{-\frac{(\ln x-\mu)^2}{4\sigma^2}} dx$$

Where $\theta, \eta, \sigma, \mu$ are variables that determines the curve for each subsystem, see equivalences in Table 5 14, and values determined by Bouwmeester et al. (2022b) presented in Table 5 15. Note, the Thermal Subsystem and Propulsion Subsystem are missing in the database because either these subsystems are typically passive in CubeSats, embedded in other subsystems or not typically applied in this framework. The electrical interfaces between subsystems are also allocated to the major subsystems (e.g. data interfaces to CDHS, power distribution to EPS). Guidance, Navigation and Control was allocated inside the Attitude, Determination and Control Subsystem since they commonly share the same instruments (Star tracker for instance).

3. Mitigation Approaches

The reliability for an element can be increased through testing and/or redundancy. At the DSM level it can be increased also by: - increasing its constituent element's individual reliability through subsystem redundancy. - Phased deployment of "new" satellites

3.1. Redundancy

Literature suggests that the power subsystems in the one with the lowest reliability (BOUWMEESTER, LANGER, GILL, 2017b, 2017c; GUO, MONAS, GILL, 2014a), and a failure in it will be catastrophic as the mission cannot operate without energy. The extreme reliability improvement would be to fly a completely redundant power subsystem, as schematically described in Figure 5 36. This was implemented in the model as an option (based on the approach by (QAISAR, RYAN, TUTTLE, 2016)) who explore the effect on the systems engineering envelopes (cost, volume, mass) for a set of different components and its combinations. Even though, based on the developer interviews, reliability at component level is present. In the case of a CubeSat a "spare" EPS will result in an increase in mass and cost and decreased in available payload volume. The rational for implementing this subsystem level redundancy, was the availability of data at this level (as data for lower levels was not available). Thus, this approach is adequate for modelling at lower level, yet here only for quantifying effects of power subsystem (EPS) redundancy over satellite reliability and systems engineering envelopes. Redundancy of subsystem is no longer assess in the test case.

3.2. Phase deployment

The approach proposed in this work (a heuristic) is to accept a "lower" element reliability, to compensate this by deploying a cluster at the same time (obtaining the required reliability though the treatment of DSM reliability as that of parallel individual elements (i.e. CubeSats)). And to maintain the set reliability deploy new spacecraft, with initial reliability higher than those already in orbit in a manner that minimizes the launch cost and number of satellites in total. This was implemented through a: Phase-deployment, variable which enables the replenishment of satellites over the mission lifetime. Relaunch-rate: as the number of months between the launches of new satellites. Care must be taken as this number is constrained by the lead times in the integration to a launch vehicle which depends on several aspects. For practical purposes a minimum of 6 months was considering, considering discussions with launch brokers and cubesat developers. DSM-relaunch-amount: number of new satellites in the new batch.

The reliability as a function of time, over a period T for single satellite generations is schematically represented in Figure 5 37. In this figure the element initially deployed will exhibit the lowest reliability, but the new ones with higher reliability and the assessment at DSM level will yield a higher value. The goal

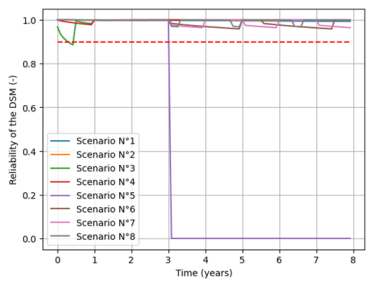

Figure 3: Results in Physics Journal cover

is to have a scenario that full fills the reliability requirements to assess the total number of satellites, and the deployment cost early in de conceptual phase.

The previous approach describes the case where only one element is deployed per batch. In the case where more than one element is deployed per batch, which actually allows to fulfil the required reliability of 0.9 of the case study example, the reliability of the system with parallel elements is given by Eq. 5 11, where R_{sat} represents the reliability of each CubeSat in the DSM and R_{DSM} the reliability of the distributed system. The more active parallel elements are present, the more chances the system will survive over time.

$$R_{DSM}\,(t) = \; 1 - \prod_{i}^{n} (1 - R_{sat,i}(t))$$

3.3. Programmed Retirement

Based on the discussion with New Space developers, and the declared lifetime of Planet satellites (Obtained from the Satellite database (UCS, 2021)), it is assumed that the CubeSats are retired after 3 years (Note for the case of ISS deployments, this might be lower as they can burn in re-entry due to atmospheric drag). This is implemented in the code as Retire-sat with True activating the state and False leaving the reliability to propagate over time. Again, a note of caution, as the reliability fits by Bouwmeester et al. can be misinterpreted as a continuum, yet experience show the CubeSats lifetime it not reaching 10 years in practice.

$$R_{sat,i}\,(t) = \{\, 0 \; < t \leq 3 \rightarrow R_{sat,i} = R_{sat,i} t > 3 \rightarrow R_{sat,i} = 0 \; [-]$$

3.4. Initial Output

The scenarios inputs are summarized in Table 5 16. Then the algorithm is run and it produces a series of results in terms of graphs and tables presented below. These include: the scenario description, the launch cost, reliability at different points in time (which is also shown graphically see Figure 5 38).

Reliability every 6 months is show in Table 5 17, where all the scenarios except 3 and 5, have a reliability above the required 0.90 all the time. Scenario

5 does not deploy new satellites, and the retirement at the end of year 3 drops reliability to 0 as no satellite is considered operational.

The deployment strategy and reliability are assessed in a Python algorithm which also provides the total number of satellites and launch cost.

4. Launch Cost Estimations

For each deployment of satellites, the cost of launch is estimated in three ways: Using only POD launchers through brokers at a cost of 150 [kUSD] for a 3 U to SSO. Being the total cost in this case the total number of satellites deployed times the cost for a brokered 3U launch. Using Space-X rideshare program at 300 kUSD for 50 kg to orbit, see Figure 5 41, makes sense from a cost perspective at least at two 3U (plus deployer POD). Using a combination of both, where the initial batch can be deployed by Space-X while the newer smaller batches (generations) used PODs. The latter due to the lower lead time for launch. If one takes the weight of a 3U cubesat by ISIS to be between 4 and 6 kg, and ISIS Quadpack deployer, capable of deploying four 3U or different configurations for a maximum of a 12 U (see Figure 5 40) with a mass between 6 and 7.5 kg one could fit up to eight triple unit CubeSats in one 50 kg launch (at the cost of two in pods, or even a bit more if one has some extra kilograms). This would represent a significant cost reduction in the order of $6 \times 150\,[kUSD] = 900[kUSD]$. Thus, based on the case study analysis the proposal for the combination is that the initial batch of satellites (i.e. the ones required to fulfil the coverage and temporal resolution of the mission are deployed in a Rideshare launch).

Thus, cost is given as the total launch cost $C_{launch,\ total}$ is given by the sum of the costs of each i-th generation in Eq. 5 12.

$$C_{launch,total} = \sum_i C_{launch,i}$$

5. Results and Discussions

The main results of this algorithm, for the purposes of analysis is the number of spacecrafts that are active over time (considering the retirement), see Table 5 18. Our mission requires to have 8 satellites operating to accomplish the temporal resolution required.

The effect of the rideshare program in the overall costs is significant. Particularly for deployments of eight 3U CubeSats (where the cost "saving" are 900 kUSD). This is the case for the initial generation of all scenarios but 3 and 4. One should keep in mind the lead times for Space-X launches are longer (24 months) while POD launches can be achieved in a 6-month windows (this is a relevant reason, ability to react to developments) that brokered launchers offer at the time when one can use the "next available ride".

Satellites are still estimated to be more expensive than the launch, as so the number of satellites need to be minimized to keep the entire mission within budget. A relevant aspect of the deployment strategy has to do with matching

the manufacturing capabilities with the launch. Based on discussions with developers, surveys, and cost modelling manpower is a cost driver and effort to balance the workload is key for a feasible project.

5.1. Study Case results

After extensive analysis of different combinations of satellite deployments and orbital altitude, the proposed mission shall be deployed at an SSO orbit at the highest altitude compliant to debris mitigation guidelines. For our scenarios this was set at 600 km. A minimum of 8 satellites are to be operational to fulfil the temporal resolution requirement, which proved to be the mission driver (as the spatial resolution was feasible within a 3U CubeSat).

Several deployment scenarios where analysed (see Table 5 18). The minimum number of CubeSats required to fulfil the mission always with 8 operational satellites is 24. Yet scenario 6, with a new batch of 4 satellites every 6 months starting at month 36 (and an initial batch of 8) was able to reduce the total number of CubeSats to 20 (with some periods where there are less than 8 satellites due to retirement).

Taking the values of Table 5 19, and comparing with respect to the requirements in Table 5 1, which state in req. 34.13 cost to be non-recurring $<$ 10 M USD and recurring $<$ 3 M USD / year, one can rephrase this requirement in terms of total budget

$$10\,[MUSD] + 8\,[yrs] \times 3\left[\frac{MUSD}{yr}\right] = 34[MUSD]$$

Or, if the project lifetime is interpreted as the operational lifetime of the satellites 8 years, plus the time for the development of maximum 3 years, thus 11 years.

$$10\,[MUSD] + 11\,[yrs] \times 3\left[\frac{MUSD}{yr}\right] = 43[MUSD]$$

The proposed architecture is within the order of magnitude. Of course, a discussion with the stakeholders regarding the recurrent costs (here just estimated) should be done early in the mission to understand the upper bounds. Cost replicability effects and learning curves become a relevant aspect when considering a 20-24 satellite cluster (as are economies of scale when procuring the COTS components or subsystems). Many of the make or buy decisions will depend on the manpower cost alternative, which is both country dependant and technology capability (as some countries might have cheaper labour, but unqualified to develop) Note the case for FireSat II in Friedenthal and Oster, which sets a spacecraft of 150 kg and four spacecraft over two different planes. The estimated cost for such a launch is 1.5 million USD per spacecraft (SPACEFLIGHT INDUSTRIES, 2022a).

6. Conclusion

As the concept in deemed feasible, further development is required and hence it must continue its analysis and trades for the constrained preliminary architectures in the previous sections.

Acknowledgements

i love my cat.

References

[1] Sweere, S.F., Valtchanov, I., Lieu, M., Vojtekova, A., Verdugo, E., Santos-Lleo, M., Pacaud, F., Briassouli, A., Cámpora Pérez, D., 2022. Deep learning-based super-resolution and de-noising for xmm-newton images. Monthly Notices of the Royal Astronomical Society 517, 4054–4069.